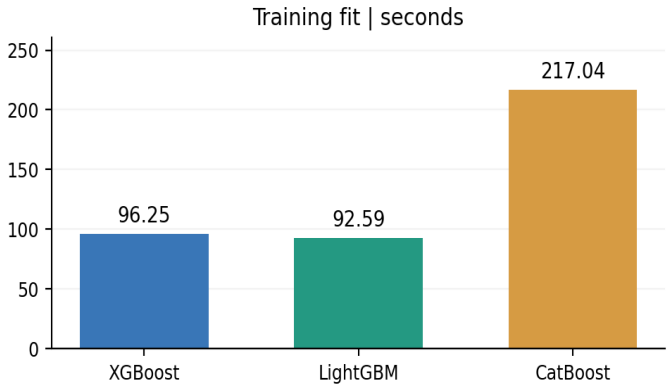
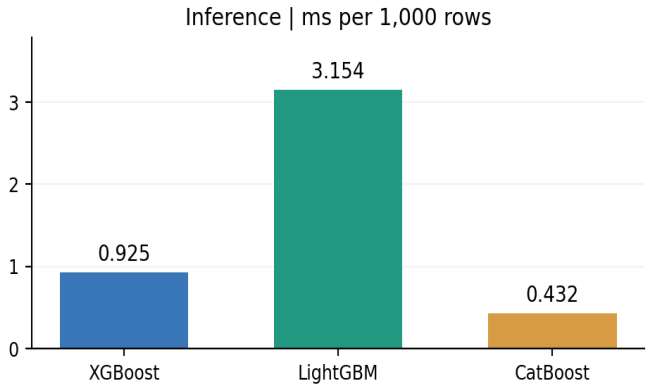
HIGGS baseline results

Three fixed 100-tree models | Test quality, training and inference

XGBoost leads prediction quality. LightGBM has the shortest recorded training time. CatBoost has the fastest measured inference.

Category	Winner	Margin against the next-best model
Test quality	**XGBoost**	ROC-AUC +0.0035; log loss 0.75% lower than LightGBM
Training time	**LightGBM**	3.66 s less than XGBoost; 3.80% shorter
Inference time	**CatBoost**	2.14x XGBoost's throughput at this batch size; 53.3% less time





Model	Test ROC-AUC Higher is better	Test log loss Lower is better	Training seconds	Inference ms / 1,000
XGBoost	0.8135	0.5256	96.25	0.925
LightGBM	0.8100	0.5296	92.59	3.154
CatBoost	0.8008	0.5421	217.04	0.432

Main interpretation

Training: LightGBM and XGBoost are close: 92.59 versus 96.25 seconds. This is a modest observed advantage. CatBoost takes 217.04 seconds, about 2.34 times LightGBM's time.

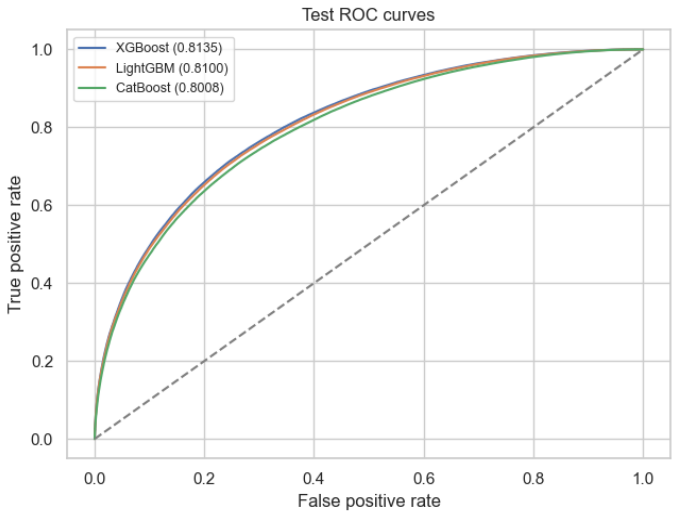
Inference: CatBoost takes 0.432 ms per 1,000 rows, versus 0.925 for XGBoost and 3.154 for LightGBM. LightGBM takes 7.30 times CatBoost's prediction time in this benchmark.

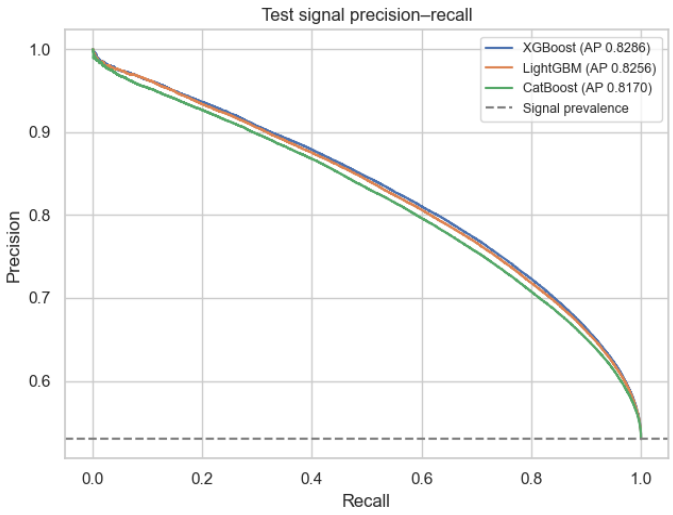
Selection: XGBoost was selected by validation log loss before test evaluation and also leads test ROC-AUC, AP and log loss. No single model wins every quality and speed measure.

Timing basis: original single-run fits; warmed-up inference on the same resident 10,000-row NumPy batch, nine repetitions, median scaled to 1,000 rows. Four prediction CPU threads. Model loading and disk reads are excluded from inference.

Quality and scope

How to interpret the result





Model	Signal AP	Signal F1	Accuracy	Brier score
XGBoost	0.8286	0.7652	71.48%	0.1763
LightGBM	0.8256	0.7633	71.20%	0.1779
CatBoost	0.8170	0.7571	70.43%	0.1827

What the metrics mean

ROC-AUC measures signal-versus-background ranking across thresholds. **Log loss** measures probability quality and penalizes confident mistakes. **AP** summarizes signal precision-recall performance; it is not trapezoidal PR-AUC. Lower Brier score indicates lower squared probability error.

Signal F1 and accuracy use thresholds selected on validation data to maximize signal F1. Thresholds were fixed before test evaluation. XGBoost's ROC-AUC advantage over CatBoost is **0.0127**, while the gap over LightGBM is **0.0035**.

Experimental scope

Validation and test each contain **500,000 events** with 28 numerical predictors. Test signal prevalence is approximately **52.99%**. The split is stratified and random across converted TFRecord records; it is not claimed to reproduce the original UCI test ordering.

All three are fixed 100-tree baselines, without Optuna or early stopping. The stated configurations use learning rate 0.1 and maximum depth 6; LightGBM uses 31 leaves. Equal tree counts do not imply identical tree structures, computation or model capacity.

Practical conclusion

XGBoost offers the strongest observed quality with training time close to LightGBM. **LightGBM** has the fastest recorded fit, but its small training-time lead needs repeated fits to establish stability. **CatBoost** favors inference speed while sacrificing quality and fitting speed under these configurations.

No confidence intervals or repeated training runs are reported. Inference timings describe warmed-up batch prediction, not single-request latency. Device settings should be verified before presenting this as a strictly CPU-only comparison.

Source: executed tables and curves in 02_higgs_evaluation.ipynb. Results are reproduced without retraining models. Relative margins use the displayed notebook values.